Axel Gomez

axel.gomez.dev@gmail.com
Axel Gomez AxelGomez16 Web profile
Queretaro, Mexico — Open to Relocation

Professional Summary

Algorithms Engineer with **4+ years in automotive R&D**, focused on **Python validation automation**, regression checks, and reporting. Reduced radar-model validation turnaround from approximately **five working days to 50 minutes** through pipeline automation and architecture refactoring. Also builds and supports a subscription product for automated news publishing.

Professional Experience

Aumovio (formerly Continental R&D) June 2022 - Present
Algorithms Engineer *Queretaro, Mexico*

- **Radar Model Validation:** Proposed and built a **Python** pipeline for data export, processing, metrics, and reports, reducing a multi-engineer validation workflow from approximately **five working days to one day**.
- Then refactored the validation architecture and migrated the codebase, bringing automated turnaround to **50 minutes with GitHub Actions** for General Motors deliverables; now extending checks to each commit and pull request.
- **Algorithm Orchestration:** Helped migrate legacy **C++** scheduling for parallel algorithm flows and per-frame failure handling.
- **LiDAR Perception & Evaluation:** Automated ground-truth evaluation and commit/pull-request checks for static-object detection and bounding-box query, surfacing performance regressions before integration.
- **Perception Integration:** Established two-week checks of combined algorithm performance; built **SQLite/Dash/Plotly** dashboards and Jaguar Land Rover reports to inform release decisions.
- **Team Enablement & Code Quality:** Set up Linux environments, trained teammates on internal tools, and helped resolve QAC findings across algorithm components under AUTOSAR/MISRA.

Postealo - Independent Software Product December 2024 - Present
Developer & Maintainer (alongside full-time employment) *Queretaro, Mexico*

- Built a **Python** workflow connecting bot intake, **LLM-assisted** text preparation, scheduling, and delivery across a news outlet's website, Facebook, Instagram, X, and WhatsApp.
- Maintain the product for a paying news organization under a **monthly subscription**, translating requests into features and providing ongoing support.

Continental Talent Entry Program (CTEP) March 2022 - June 2022
Systems Engineer Trainee *Queretaro, Mexico*

- Completed requirements, development, and testing work on a door-lock and window-control vehicle prototype before transitioning to algorithms engineering.

Technical Skills

Programming & Scripting: Python, C++, Bash
Workflow Automation: Data export/processing, report generation, commit/pull-request validation
CI/CD & Platforms: GitHub Actions, Jenkins, Git, Linux, Docker
Data & Dashboards: SQLite, Dash, Plotly, Pandas, NumPy, Matplotlib, ground-truth metrics
Spoken Languages: English, Spanish

Education

Master of Science in Computer Science (Artificial Intelligence) Expected July 2027
Autonomous University of Queretaro *Online, 2025 - Present*

Bachelor of Science in Engineering Physics 2017 - 2021
Autonomous University of Queretaro *Queretaro, Mexico*